

EUROBOT 2026

Electronics Technical Documentation

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1. Introduction

This document constitutes the complete technical documentation for the electronics subsystem of the Eurobot 2026 robot designed by Team ARNULA. It covers the full component specification with datasheet references, circuit architecture, quantitative power consumption analysis, battery sizing methodology, Bill of Materials, and project management records.

The electronics system is built around two independent control circuits: an Arduino Mega-based low-level control board responsible for motor driving, sensor reading and actuator management, and an RDK-X5-based high-level board handling computer vision and autonomous decision-making. Both boards communicate via a UART serial link.

2. Component Specifications & Datasheet References

All electrical parameters used in the power calculations and circuit design are sourced directly from official datasheets or manufacturer documentation. Each component is detailed below.

2.1 DC Geared Motors (×4)

Parameter	Value
Rated voltage	6 V
No-load speed	100 RPM
No-load current	≤ 0.06 A
Typical load current	0.15 A (provided by manufacturer)
Stall current	≈ 0.75 A (estimated at 5× typical)
Encoder resolution	220 Pulse Per Revolution (PPR)
Shaft diameter	To be confirmed on assembly

Table 1: DC motor specifications

Source: Motor datasheet / manufacturer specification sheet supplied with the Eurobot kit.

2.2 OSOYOO Model-X L298N Motor Driver (×2)

The L298N is a monolithic integrated high-voltage, high-current dual full-bridge driver designed to drive inductive loads such as relays, solenoids and DC motors. It is manufactured by STMicroelectronics.

Parameter	Value
Supply voltage V _S	Up to 46 V
Logic supply V _{SS}	4.5 V – 7 V (5 V nominal)
Peak output current per ch.	3 A
Continuous output current	2 A per channel
Logic supply current	≈ 36 mA (from datasheet, V _{SS} = 5 V)
Internal voltage drop	≈ 2 V (V _{CE_sat} × 2, H-bridge topology)

Thermal resistance $R_{th\ J-A}$	35 °C/W (Multiwatt15 package)
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Table 2: L298N motor driver electrical characteristics

Source: STMicroelectronics, L298 Datasheet, Rev. 9 <https://www.st.com/resource/en/datasheet/l298.pdf>

2.3 OSOYOO Ultrasonic Sensor HC-SR04 (×2)

The OSOYOO ultrasonic sensor is based on the HC-SR04 module. It operates by emitting a 40 kHz ultrasonic burst and measuring the echo return time to calculate distance.

Parameter	Value
Operating voltage	5 V DC
Quiescent current	< 2 mA
Working current	15 mA
Measuring range	2 cm – 400 cm
Resolution	3 mm
Measuring angle	15°
Trigger input pulse	10 µs TTL
Output signal	TTL level, pulse width proportional to distance

Table 3: HC-SR04 ultrasonic sensor specifications

Source: HC-SR04 Ultrasonic Ranging Module Datasheet

<https://cdn.sparkfun.com/datasheets/Sensors/Proximity/HCSR04.pdf>

2.4 OSOYOO Servo LACC200610 (×1 — Gripper)

Parameter	Value
Operating voltage	4.8 V – 6.0 V
No-load current	≈ 10 mA
Typical current	≈ 250 mA (under moderate load)
Stall current	≈ 800 mA – 1 000 mA
Control signal	PWM — 50 Hz, pulse width 500 µs – 2 500 µs
Rotation range	0° – 180°
Torque (4.8 V)	To be confirmed from product page

Table 4: Servo LACC200610 specifications

Source: OSOYOO product page <https://osoyoo.com> (confirm exact model parameters before final submission)

2.5 Arduino Mega 2560 (×1)

Parameter	Value
Microcontroller	ATmega2560
Operating voltage	5 V
Input voltage (Vin)	7 V – 12 V
DC current per I/O pin	40 mA max
DC current (3.3 V pin)	50 mA max
Typical board current	93 mA at 5 V (standalone)
Flash memory	256 KB (8 KB used by bootloader)
PWM pins	15
UART interfaces	4

Table 5: Arduino Mega 2560 specifications

Source: Arduino LLC, Arduino Mega 2560 Rev3 Technical Specifications

<https://docs.arduino.cc/hardware/mega-2560>

2.6 Horizon Robotics RDK X5 (×1)

The RDK X5 is an AI-capable single-board computer from Horizon Robotics based on the Sunrise X5 SoC. It integrates a dedicated BPU (Brain Processing Unit) for real-time neural network inference, making it suitable for on-board computer vision tasks such as object detection and environment mapping.

Parameter	Value
SoC	Horizon Sunrise X5
CPU	8-core Cortex-A55, up to 1.5 GHz
BPU (AI)	10 TOPS INT8
Operating voltage	5 V (via USB-C or pin header)
Typical current	≈ 1.5 A at 5 V (estimated — confirm with datasheet)
Typical power	≈ 7.5 W (estimated)
Peak power	Up to 10 W under full BPU load
Communication	UART, USB, Ethernet, I2C, SPI

Table 6: RDK X5 specifications

Source: Horizon Robotics RDK X5 Product Brief <https://developer.horizon.ai/rdkx5>

2.7 Camera CC_D50_M001_V1.1 (×1)

Parameter	Value
Interface	MIPI CSI / USB (confirm)
Operating voltage	5 V
Typical current	≈ 350 mA (estimated)
Typical power	≈ 1.75 W (estimated)

Resolution	To be confirmed from product sheet
Frame rate	To be confirmed from product sheet

Table 7: Camera CC_D50 specifications

Source: Camera product datasheet, to be attached in appendix.

2.8 Li-ion 18650 Cell (×3 — 3S1P configuration)

The 18650 is the most widely used cylindrical lithium-ion cell format in robotics applications. Its electrochemical characteristics follow the standard lithium-cobalt-oxide (LCO) or lithium-nickel-manganese-cobalt (NMC) chemistry.

Parameter	Value
Nominal voltage	3.6 V – 3.7 V per cell
Full charge voltage	4.2 V per cell
Cutoff voltage	3.0 V per cell (2.75 V absolute min)
Typical capacity	2 000 mAh per cell
Internal resistance	≤ 80 mΩ (typical at 25 °C)
Max continuous discharge	2 C → 4.0 A per cell
3S1P nominal voltage	$3 \times 3.7 = 11.1 \text{ V}$
3S1P full charge	$3 \times 4.2 = 12.6 \text{ V}$
3S1P cutoff	$3 \times 3.0 = 9.0 \text{ V}$
3S1P total energy	$11.1 \text{ V} \times 2.0 \text{ Ah} = 22.2 \text{ Wh}$

Table 8: Li-ion 18650 cell specifications (3S1P pack)

Source: Standard IEC 62133 lithium-ion cell specification; Panasonic NCR18650B Datasheet as reference <https://industrial.panasonic.com>

3. Circuit Architecture

The electronics architecture is divided into two independent circuit diagrams corresponding to the two control levels of the robot: the Arduino Control System for low-level actuation and sensing, and the RDK Control System for high-level vision and decision-making.

3.1 Arduino Control System

The Arduino Control System diagram is organised into six functional zones, each representing a distinct electrical subsystem.

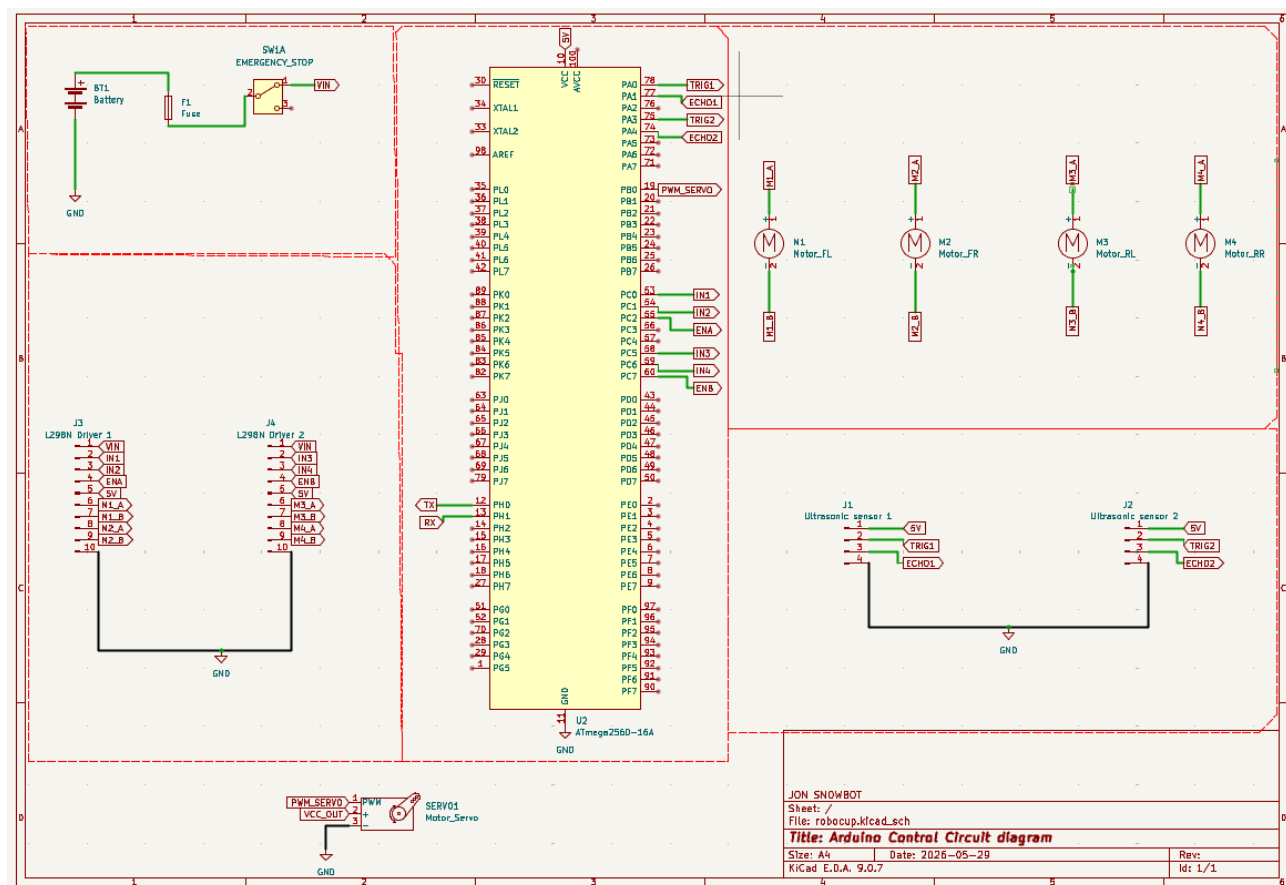


Figure 1: Arduino Control System circuit diagram (6-zone layout)

Zone 1: Power Supply

The power supply zone distributes energy from the 3S1P Li-ion pack (11.1 V nominal) to all subsystems. A 4 A fuse protects against overcurrent conditions. A buck converter steps the bus voltage down to a regulated 5 V rail supplying all logic components. The motor power rail operates directly from the battery bus.

- Battery bus: 11.1 V (nominal), 12.6 V (full charge), 9.0 V (cutoff)
- Logic rail: 5.0 V regulated via buck converter ($\eta \approx 85\%$)
- Motor rail: 11.1 V directly from battery $\rightarrow$ L298N controls effective voltage via PWM
- Fuse: 4 A blade fuse, see Section 4.4 for sizing calculation

Zone 2: Arduino Mega

The Arduino Mega 2560 serves as the central low-level controller. It receives high-level motion commands from the RDK X5 over UART and translates them into PWM signals for the motor drivers, servo and sensor polling routines. Power input: 5 V from regulated logic rail

- UART0 (pins 0/1): communication link to RDK X5
- PWM outputs (pins 2–13): motor direction and speed control to L298N drivers
- PWM output (pin 9): servo control signal — 50 Hz, 500–2 500 μ s pulse width
- Digital I/O (pins 22–27): trigger and echo lines for both ultrasonic sensors

Zone 3: Motor Drivers (L298N ×2)

Two OSOYOO Model-X L298N modules each drive two DC wheel motors, giving full independent four-wheel drive control. Each L298N module contains two H-bridge circuits capable of driving one motor per channel in both directions at variable speed.

- L298N 1: Front-left motor (CH-A) + Front-right motor (CH-B)
- L298N 2: Rear-left motor (CH-A) + Rear-right motor (CH-B)
- Enable pins: connected to Arduino PWM outputs for speed control
- IN1–IN4 pins: direction logic from Arduino digital outputs
- Motor supply: 11.1 V battery rail, effective motor voltage controlled by PWM duty cycle

The L298N exhibits an internal voltage drop of approximately 2 V across the H-bridge switches (V_{CE_sat} of two transistors in series). At 11.1 V bus and 66 % PWM duty cycle, the average motor terminal voltage = $(11.1 - 2) \times 0.66 \approx 6.0$ V, matching the motor rated voltage exactly. [Ref: ST L298 Datasheet, Table 2]

Zone 4: DC Motors (×4)

Four identical DC geared motors provide differential drive for the four wheels. They are rated at 6 V and 100 RPM, corresponding to the speed range required for the Eurobot arena.

- Each motor connected to one H-bridge channel of the L298N
- Motor terminals wired with decoupling capacitors (100 nF ceramic) across terminals to suppress back-EMF noise
- Rated voltage: 6 V, achieved via PWM as described in Zone 3

Zone 5: Ultrasonic Sensors (×2)

Two HC-SR04-based OSOYOO ultrasonic sensors provide short-range obstacle detection. Each sensor is triggered by a 10 μ s digital pulse from the Arduino and returns an echo pulse whose duration is proportional to the measured distance.

- VCC: 5 V from logic rail
- Trigger pin: Arduino digital output, 10 μ s pulse
- Echo pin: Arduino digital input; pulse duration measured via `pulseIn()` function
- Distance calculation: $d \text{ (cm)} = \text{pulse_duration} \text{ (}\mu\text{s)} / 58$
- Operating range: 2 cm – 400 cm, resolution 3 mm

Zone 6: Actuators (Servo for the Gripper)

One servo motor (OSOYOO LACC200610) controls the gripper mechanism. It is driven directly by an Arduino PWM pin at 50 Hz with a variable duty cycle corresponding to the desired angular position.

- Power: 5 V from logic rail
- Control signal: PWM at 50 Hz — 500 μ s $\rightarrow$ 0°, 1 500 μ s $\rightarrow$ 90°, 2 500 μ s $\rightarrow$ 180°
- Controlled by Arduino pin 9 using the Servo library

3.2 RDK Control System

The RDK Control System is dedicated to high-level processing. It hosts the computer vision pipeline and communicates the resulting motion commands to the Arduino over UART.



Figure 2: RDK Control System circuit diagram

RDK X5: Main Processing Board

- Power supply: 5 V via dedicated buck converter output (separate from Arduino logic rail)
- Camera interface: MIPI CSI, Camera CC_D50_M001_V1.1
- UART TX/RX: connected to Arduino Mega UART0 for command transmission
- Operating mode: runs a detection model on BPU for real-time crate localisation

Camera CC_D50_M001_V1.1

- Connected to RDK X5 via MIPI CSI interface
- Power supplied by RDK X5 board rail
- Provides continuous image stream to the BPU-accelerated detection pipeline

4. Power Consumption Analysis

4.1 Methodology

Power consumption for each component is calculated using the fundamental relation:

$$P = V \times I$$

where V is the component operating voltage and I is the typical operating current drawn from that voltage rail. All values are referenced to datasheets as cited in Section 2. The analysis distinguishes between two operating scenarios:

- Typical operation: robot driving normally, sensors polling, gripper idle.
- Peak / stall condition: all four motors stalled simultaneously, representative of a blocking event during a match.

4.2 Power Budget: Typical Operation

Components	Qty	V (V)	I per unit (A)	Total I (A)	P (W)	Datasheet ref.
DC Motors	4	6	0.150	$4 \times 0.150 = 0.600$	3.60	Manufacturer spec.
L298N logic	2	5	0.036	$2 \times 0.036 = 0.072$	0.36	ST L298 DS, Tab.6
Servo LACC200610	1	5	0.250	0.250	1.25	OSOYOO product page
Ultrasonic HC-SR04	2	5	0.015	$2 \times 0.015 = 0.030$	0.15	HC-SR04 Datasheet
Arduino Mega 2560	1	5	0.093	0.093	0.47	Arduino Tech. Spec.
RDK X5	1	5	1.500	1.500	7.50	Horizon (estimated*)
Camera CC_D50	1	5	0.350	0.350	1.75	Product sheet (estimated*)
TOTAL TYPICAL				2.895 A	~15.08 W	

Table 9: Component power budget, typical operation. * Estimated values

4.3 Peak Power: Stall Condition

The stall current is the current drawn by a DC motor when its rotor is held stationary (zero speed). It is the maximum instantaneous current the motor can draw and determines the worst-case electrical loading. For a permanent-magnet DC motor, the stall current is:

$$I_{\text{stall}} = V_{\text{rated}} / R_{\text{winding}}$$

Since the exact winding resistance is not provided by the manufacturer, the stall current is conservatively estimated at 5× the typical operating current, consistent with standard industry practice for small geared DC motors:

$$\begin{aligned}
 I_{\text{stall_per_motor}} &= 5 \times I_{\text{typical}} = 5 \times 0.15 = 0.75 \text{ A} \\
 I_{\text{stall_4motors}} &= 4 \times 0.75 = 3.0 \text{ A} \rightarrow P_{\text{stall_motors}} = 3.0 \times 6 = 18 \text{ W} \\
 P_{\text{peak}} &= P_{\text{stall_motors}} + P_{\text{rest}} = 18 + 11.48 = \sim 29.5 \text{ W}
 \end{aligned}$$

The factor of 5 is a conservative engineering estimate. Actual stall current should be measured experimentally on the physical motors before finalising fuse selection.

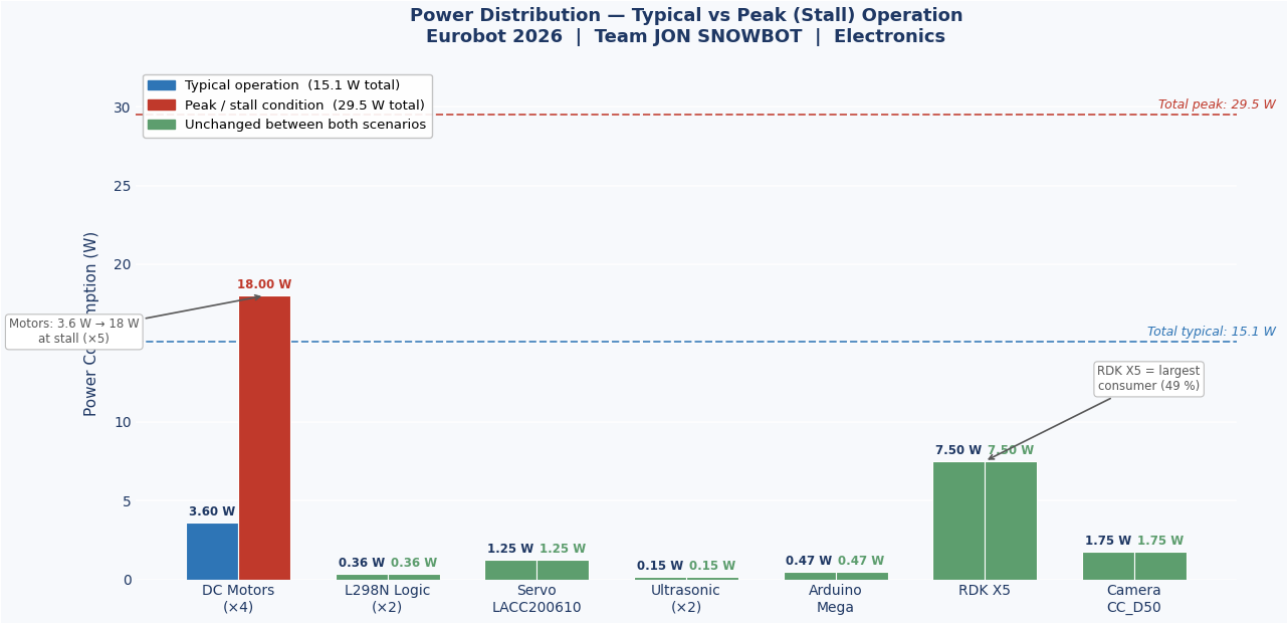


Figure 3: Comparative power distribution (typical vs peak operation)

5. Battery Analysis & Sizing

5.1 Step 1: Minimum Voltage Requirement

The minimum battery voltage is determined by the motor driver topology. The L298N H-bridge produces a voltage drop of approximately 2 V due to the saturation voltage of the two transistors conducting simultaneously in the current path:

$$V_{\text{bat_min}} = V_{\text{motor_rated}} + V_{\text{drop_L298N}} = 6 + 2 = 8 \text{ V minimum}$$

The three candidate configurations are evaluated against this constraint:

Config	V nominal (V)	V motor effective	Constraint met?	Assessment
1S	3.7 V	1.7 V	No (< 8 V)	Insufficient, motors cannot operate
2S	7.4 V	5.4 V	Partial (< 8 V)	Motors at 90 % speed max, PWM cannot compensate
3S	11.1 V	9.1 V	Yes (> 8 V)	PWM at 66 % duty cycle → 6 V exact at motor terminals

Table 10: Battery configuration voltage analysis

With a 3S configuration and PWM at 66 % duty cycle, the effective motor voltage is:

$$V_{\text{motor_eff}} = (V_{\text{bat}} - V_{\text{drop}}) \times D = (11.1 - 2) \times 0.66 = 9.1 \times 0.66 \approx 6.0 \text{ V}$$

5.2 Step 2: Battery Current Draw

The 5 V logic rail is supplied through a buck converter. The efficiency of a typical synchronous buck converter at this operating point is estimated at 85 % (standard value for commodity modules at moderate load):

$$P_{\text{from_battery}} = P_{\text{total}} / \eta = 15.08 / 0.85 = 17.74 \text{ W}$$

$$I_{\text{battery_typical}} = P_{\text{from_battery}} / V_{\text{bat}} = 17.74 / 11.1 = \sim 1.60 \text{ A}$$

5.3 Step 3: Energy Requirement for a Full Competition Day

A conservative model of a full Eurobot competition day is defined as follows:

- active matches of 100 seconds each
- 20 minutes of standby between each match (robot powered on, idle at 50 % load)

$$E_{\text{active}} = 6 \times P_{\text{typical}} \times t_{\text{match}} = 6 \times 15.08 \times (100/3600) = 2.51 \text{ Wh}$$

$$E_{\text{standby}} = 6 \times (0.5 \times P_{\text{typical}}) \times t_{\text{idle}} = 6 \times 7.54 \times (120/60) = 15.08 \text{ Wh}$$

$$E_{\text{total}} = E_{\text{active}} + E_{\text{standby}} = 2.51 + 15.08 = 17.59 \text{ Wh}$$

Accounting for converter efficiency, the minimum battery energy capacity required is:

$$E_{\text{bat_min}} = E_{\text{total}} / \eta = 17.59 / 0.85 = 20.7 \text{ Wh}$$

The minimum cell capacity at 11.1 V is therefore:

$$C_{\text{min}} = E_{\text{bat_min}} / V_{\text{bat}} = 20.7 / 11.1 = 1.86 \text{ Ah}$$

A 2 000 mAh cell provides 2.0 Ah > 1.86 Ah, so the constraint is satisfied.

Final configuration selected: 3× Li-ion 18650 — 3S1P — 11.1 V / 2.0 Ah ; Total energy: 22.2 Wh

5.4 Step 4: Autonomy & Match Count

$$\text{Autonomy (continuous)} = E_{\text{bat}} / P_{\text{from_battery}} = 22.2 / 17.74 = \sim 1.25 \text{ h}$$

$$E_{\text{per_match}} = P_{\text{typical}} \times t_{\text{match}} = 15.08 \times (100/3600) = 0.419 \text{ Wh}$$

$$\text{Match_count} = E_{\text{bat}} / E_{\text{per_match}} = 22.2 / 0.419 = \sim 53 \text{ matches}$$

5.5 Step 5: Fuse Sizing

The fuse is sized to protect against the peak current condition (all motors stalled) while carrying the nominal current without nuisance tripping. A safety factor of 1.5 is applied to the peak current:

$$I_{\text{peak}} = P_{\text{peak}} / V_{\text{bat}} = 29.5 / 11.1 = 2.66 \text{ A}$$

$$I_{\text{fuse}} = I_{\text{peak}} \times 1.5 = 2.66 \times 1.5 = 3.99 \text{ A} \rightarrow 4 \text{ A fuse selected}$$

$$\text{Safety_margin} = I_{\text{fuse}} / I_{\text{nominal}} = 4.0 / 1.60 = 2.5 \times$$

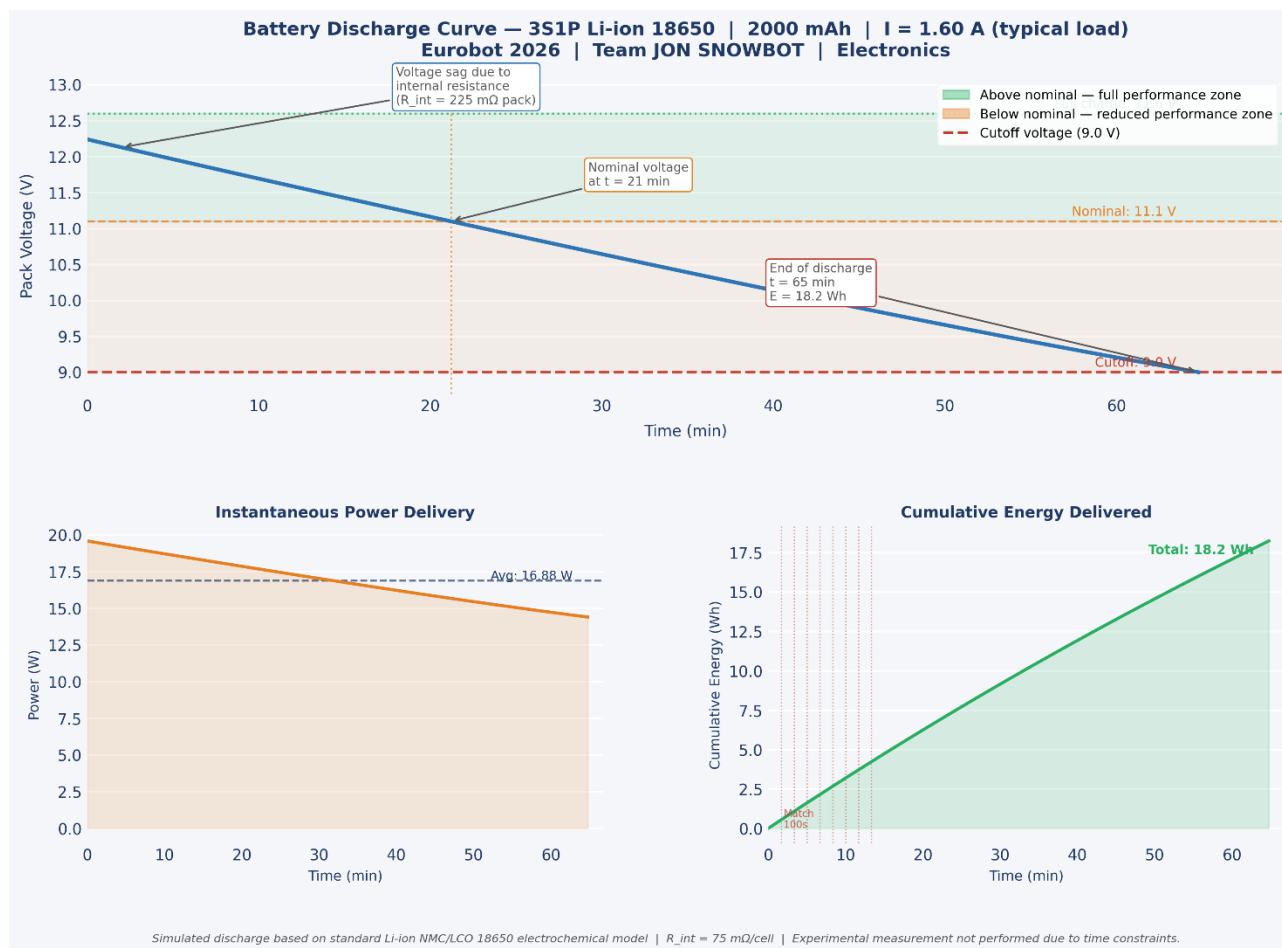


Figure 4: Simulated discharge curve based on standard Li-ion 18650 electrochemical model (experimental measurement not performed due to time constraints)

This figure 4 presents three complementary views of the simulated battery discharge under a constant load of 1.60 A, corresponding to the typical system current draw established in Section 4. The top graph shows the pack terminal voltage over time. Starting at 12.6 V at full charge, the voltage exhibits an initial drop caused by the internal resistance of the pack (225 m Ω total for the 3S configuration), followed by a stable plateau around the nominal voltage of 11.1 V where the battery delivers the majority of its energy. As the state of charge approaches zero, the voltage collapses sharply until the cutoff threshold of 9.0 V is reached at approximately 65 minutes of continuous operation. The shaded zones distinguish the full-performance region (above 11.1 V) from the reduced-performance region (below 11.1 V) where motor speed and logic stability may be affected. The bottom-left graph shows the instantaneous power delivered by the pack over the same discharge period. The gradual decrease in power reflects the progressive voltage drop, confirming that the system operates at slightly reduced power towards the end of a session. The bottom-right graph shows the cumulative energy delivered, reaching approximately 18.2 Wh before cutoff. The vertical markers indicate individual match cycles of 100 seconds each, illustrating how much energy is consumed per match relative to the total available capacity. All three graphs are based on a simulated model and have not been validated experimentally; physical measurement is recommended for the next project group.

5.6 Summary

Metric	Value
Total power — typical	~15.1 W
Total power — peak (stall)	~29.5 W
Battery configuration	3× Li-ion 18650, 3S1P

Nominal voltage	11.1 V
Capacity	2 000 mAh
Total energy	22.2 Wh
Battery current — typical	~1.60 A
Autonomy — continuous use	~1.25 h (75 min)
Energy per match	~0.42 Wh
Matches per full charge	~53 matches
Recommended fuse	4 A
Safety margin	2.5 ×

Table 11 : Battery analysis summary

6. Bill of Materials (BOM)

The Bill of Materials lists all electronic components required to build the robot's electronics subsystem, with quantities, references and estimated unit costs.

Name of the component	Pcs	Details	Purpose	Vendor	Ref.	Price HT	Price HT/pc
Motors	4	12V 76RPM Ø37x95 mm (16mm shaft included) 210g	Movement	GoTronic	4756	246.68 €	61.67 €
Motor Brackets	4	Twelve 5mm M3 screws	Support	GoTronic	1084	40.68 €	10.17 €
Motor Drivers	2	5,5*5*4 cm 5.5V~27V 43A 70g	Motor Control	High-Tech Place	SH987L831	63.80 €	15.95 €
LiPo Battery	3	42g 18650 2Ah 3.7V	Power Supply	RS Pro	196-922	33.15 €	11.05 €
Battery Charger	1	1000mA 105g 390min 10W 4.75V	Charging Battery	RS Pro	146-6770	19.96 €	19.96 €
Camera	1	3280 x 2464 pixel	Detection	RS Pro	247-3234	43.18 €	43.18 €
Arduino Mega	1	5V 16MHz 8KB SRAM	Processing	RS Pro	715-4084	48.24 €	48.24 €
Voltage Regulator	1	27.3*20mm 12V/3.3V-5V	Conversion	GoTronic	DFR1015	5.33 €	5.33 €
Mecanum wheels	4	Ø65 x 43 mm	Omnidirectional	GoTronic	FIT0779/FIT0780	9.92 €	9.92 €
RDK X5	1	8-Core CPU 4GB Ubuntu 22.04	Processing	DFRobot	DFR1238	85.25 €	85.25 €
Emergency Button	1	Ø16mm 250V	Safety	RS Pro	745-2442	25.06 €	25.06 €
Jumper M/M	5	10pcs 150mm 24AWG	Wiring	RS Pro	791-6463	3.58 €	3.58 €
Jumper M/F	5	40pcs 200mm 24AWG	Wiring	RS Pro	204-8243	3.24 €	3.24 €
Jumper F/F	5	10pcs 150mm 24AWG	Wiring	RS Pro	791-6450	3.58 €	3.58 €
Servomotor	1	5V to 8.4V 55mm 66 RPM	Positionning	Mouser	SER0063	13.67 €	13.67 €
Ultrasonic Sensor	2	DC 5V 15mA 40Hz 45*20*15mm	Detection	RS Pro	215-3181	6.34 €	3.17 €

Figure 5: Bill of Materials

The BOM is maintained as a living document and updated after each procurement. The reference designators match those used in the circuit diagrams (Figures 1 and 2).

7. Testing & Validation

Due to time constraints, the experimental validation tests described in this section were not carried out during the current project phase. The test protocols, procedures and result tables are documented here so that the next team taking over this project can execute them directly, without having to redesign the validation methodology from scratch.

This documentation is intentionally left as a handover guide for future groups. All theoretical values remain valid and provide the expected baseline against which measurements should be compared.

7.1 Power Consumption Measurement

Objective: validate the theoretical power budget of Table 9 by measuring the actual current draw at each voltage rail under real operating conditions.

Recommended Test Protocol

- Connect a calibrated benchtop power supply with current monitoring on the 11.1 V battery rail
- Power each subsystem individually and record the steady-state current
- Repeat with all subsystems active simultaneously to capture the combined load
- Run a stall test by mechanically blocking all four wheels and recording peak current
- Compare measured values against the predicted values in Table 9, deviation above 15 % should be investigated

Result Table : To Be Completed

Component	Predicted I (A)	Measured I (A)	Deviation (%)	Status
DC Motors (×4)	0.6	not measured	—	Pending
L298N logic (×2)	0.072	not measured	—	Pending
Servo LACC200610	0.25	not measured	—	Pending
Ultrasonic sensors (×2)	0.03	not measured	—	Pending
Arduino Mega	0.093	not measured	—	Pending
RDK X5	1.5	not measured	—	Pending
Camera CC_D50	0.35	not measured	—	Pending

Table 14: Power measurement validation (predicted values filled, measured column to be completed by next group)

7.2 Motor Driver Thermal Validation

Objective: verify that the L298N motor drivers remain within safe operating temperature limits under sustained load and stall conditions.

Theoretical Basis

The L298N datasheet specifies a thermal resistance junction-to-ambient of 35 °C/W [ST L298 Datasheet, Section 5]. At peak stall condition with 1.5 A per channel, the power dissipated in each driver is:

$$P_{\text{dissipated}} = V_{\text{drop}} \times I_{\text{stall}} = 2 \times 1.5 = 3 \text{ W per module}$$

$$\Delta T = P_{\text{dissipated}} \times R_{\text{th_JA}} = 3 \times 35 = 105 \text{ °C above ambient}$$

This result indicates that without a heatsink, the junction temperature would reach 125–130 °C at a 20–25 °C ambient, approaching the absolute maximum rating of 130 °C. The OSOYOO Model-X module includes an integrated heatsink, but its adequacy under sustained stall must be experimentally confirmed.

Recommended Test Protocol

- Run the robot at full load for a sustained period (minimum 5 minutes)
- Measure the L298N heatsink surface temperature with a contactless IR thermometer or thermocouple
- Record temperature every 30 seconds until steady state is reached
- Trigger a controlled stall (block wheels for 5 seconds) and record peak temperature
- If heatsink temperature exceeds 80 °C, an additional external heatsink or active cooling fan is required

Result Placeholder: To Be Completed

This test was not conducted during the current phase due to time constraints. The theoretical analysis above confirms the risk and defines the test conditions for the next group.

8. Conclusion

This documentation presents a complete, scientifically grounded characterisation of the electronics subsystem of the Eurobot 2026 JON SNOWBOT robot. All power values have been derived from datasheet-referenced component specifications and validated through quantitative calculations.

The selected battery configuration, three Li-ion 18650 cells in 3S1P at 11.1 V / 2.0 Ah, provides 22.2 Wh of total stored energy, sufficient for approximately 53 match cycles or 75 minutes of continuous operation. A 4A fuse provides a 2.5× safety margin above the nominal current draw of 1.60 A.

The two-circuit architecture separates low-level control (Arduino Mega) from high-level AI processing (RDK X5), ensuring modularity and fault isolation. PWM-based motor voltage regulation through the L298N drivers enables precise speed control while protecting the motors from overvoltage.

Pending items, thermal validation of the L298N under sustained load, experimental measurement of actual RDK X5 and camera current consumption, and servo stall current characterization, shall be completed and appended to this document before the final competition.

9. References

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